

FINAL PROJECT PROPOSAL

SentinelSwarm — Agentic AI for Autonomous Security Operations

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The Arzens | BS Artificial Intelligence — AI Automation & Security Engineering

1. PROJECT TITLE

SentinelSwarm — An Explainable Multi-Agent AI System for Autonomous Security Operations.

2. PROBLEM STATEMENT

Security Operations Center (SOC) teams are overwhelmed by alert volume. Analysts spend the majority of their time on repetitive triage — reading logs, checking indicators, deciding severity — rather than making high-value judgment calls. Traditional SIEM/SOAR tools automate rules, not reasoning; they cannot investigate an alert the way a human analyst would, by pivoting across data sources, cross-referencing threat intelligence, and building a coherent narrative of what happened and why it matters.

3. PROPOSED SOLUTION

SentinelSwarm is a multi-agent AI system that functions as a virtual SOC analyst. A dedicated AI agent will ingest security alerts and logs, reason about their severity using a Large Language Model, enrich its analysis with real threat-intelligence data, and generate a clear, human-readable explanation for every decision it makes — allowing a human reviewer to approve, override, or learn from the agent's reasoning rather than blindly trusting a black-box score.

4. PROJECT OBJECTIVES

- Automate the initial triage and severity classification of security alerts.
- Reduce analyst workload by filtering noise and prioritizing genuine threats.
- Ensure every AI decision is explainable and auditable, not a black-box output.
- Build a foundation that can scale from a single agent into a full multi-agent SOC platform.

5. KEY FEATURES

- AI Triage Agent that classifies incoming alerts by severity (Low / Medium / High / Critical).
- Threat-intelligence enrichment using a live open-source feed (e.g., abuse.ch).
- Human-readable reasoning trace attached to every AI decision.
- Simple analyst dashboard: Alert → AI Reasoning → Severity → Approve/Reject.
- Modular design allowing future agents (Investigation, Response, Reporting) to be added.

6. TECHNOLOGIES & TOOLS TO BE USED

Python, LangGraph/CrewAI (agent orchestration), Claude/GPT API (LLM reasoning), Elasticsearch (log indexing), FastAPI (backend), PostgreSQL (data storage), React (dashboard interface), Docker (containerized deployment).

7. EXPECTED OUTCOME

A working, demoable AI Triage Agent that takes in raw security alert data and produces a classified, explained, and human-reviewable output end-to-end — serving as the proof-of-concept foundation for a full agentic Security Operations Center platform to be developed further as my Final Year Project.

8. CHALLENGES I MAY FACE

- LLM hallucination or inconsistent reasoning on ambiguous alerts — mitigated by strict prompt grounding and confidence scoring.
- Lack of real-world labeled data — addressed using public security datasets and synthetic test alerts.
- Balancing explainability with response speed as agent complexity grows.
- Learning multi-agent orchestration frameworks within a limited timeframe.

9. FUTURE IMPROVEMENTS

Beyond the internship, SentinelSwarm will expand into a true multi-agent platform — adding an Investigation Agent, a Response Recommendation Agent, and a Reporting Agent, each contributing to a unified, auditable incident workflow. Longer-term versions will incorporate automated digital-forensics timeline reconstruction, API security testing, and privacy-preserving zero-trust risk scoring, evolving the project into a SaaS-ready "SOC-in-a-box" platform for small and mid-sized organizations.

10. WHY I CHOSE THIS PROJECT

As a BS Artificial Intelligence student specializing in AI Automation and Security Engineering, I wanted a project that goes beyond a typical internship deliverable. Agentic AI applied to cybersecurity sits at the exact intersection of my specialization and the industry's current direction — every major security vendor is investing heavily in AI-driven security operations. This project lets me build genuine technical depth in LLM-based agents, real security operations knowledge, and a system I can realistically keep developing into my Final Year Project and eventually my own startup, while directly strengthening my profile for roles at leading AI and cybersecurity organizations.